

Polynomial Regression

Machine Learning Assignment 1 | Roll No. BT2024146

Two separate regression tasks predict the assigned target variables from their input features. Each task has 1,000 training rows and 1,000 test rows. The assignment limits total polynomial degree to 10 for var1 and 20 for var2; hidden test labels were unavailable.

Problems and data

Problem	Features	Target	Train / test	Target range	Max degree
var1	6: x1, x2, x3, x4, x5, x6	Net Power Score	1,000 / 1,000	-9.59 to 13.95	10
var2	3: x1, x2, x3	Thermal Anomaly Score	1,000 / 1,000	-29.85 to 39.75	20

Observed checks: feature range -1 to 1; 0 missing values; 0 duplicate training rows; test feature duplicates: 2 (var1), 6 (var2). Duplicate test rows were kept in their supplied order.

Validation and method

Model comparisons use shuffled RepeatedKFold cross-validation with 5 folds and 2 repeats (10 validation folds total; random seed 2026). Each fold trains on 800 rows and validates on 200. The repeated splits reduce dependence on one random partition while keeping the full permitted degree searches practical.

PolynomialFeatures expands all monomials up to each candidate total degree. Ridge regression searched degrees 1-10 for var1 and 1-20 for var2, with the configured alpha grid at every degree. Unregularized least squares was also evaluated where the expanded feature count was below 800. The one-standard-error rule selects the lowest degree within one fold-to-fold standard error of the minimum mean validation MSE.

Leakage control: polynomial expansion is deterministic, and StandardScaler is fitted only on each fold's training rows before transforming that fold's validation rows. After selection, each final pipeline is refit on all 1,000 labeled rows. MSE and R-squared are averaged over validation folds.

Results and final predictions

Problem	Selected polynomial Ridge	Terms	CV MSE	CV R-squared
var1	Degree 5, alpha 20	461	0.515157	0.951535
var2	Degree 10, alpha 1	285	0.262251	0.994186

Degree 11 achieved the lowest raw cross-validation MSE. However, degree 10 was selected using the one-standard-error rule because its validation error was within one standard error of the minimum while requiring fewer polynomial terms.

The raw minimum was 0.260706 at degree 11 (alpha 1, 363 terms). The selected degree-10 error was 0.262251, below the one-SE cutoff of 0.268001.

OLS comparison and generalization

Problem	Best OLS degree	CV MSE	CV R-squared
var1	4	0.815519	0.923266
var2	8	0.284971	0.993656

Ridge improved on the best OLS candidate for both tasks. For var1, degree 10 had lower training MSE (0.132 vs 0.200 at selected degree 5) but higher validation MSE (1.020 vs 0.515). For var2, from degree 11 to 15 training MSE fell (0.169 to 0.152) while validation MSE rose (0.261 to 0.282).

Focused Ridge alpha refinement

The local grids were var1: 1, 3, 5, 7, 10, 15, 20, 30, 50 and var2: 0.1, 0.3, 0.5, 0.7, 1, 1.5, 2, 3, 5. They used the same folds and fold-local scaling. A candidate required at least 1% lower mean CV MSE and paired mean fold improvement greater than its paired standard error to be adopted.

Problem	Baseline / best alpha	Best local MSE	Relative gain	Paired gain (SE)	Decision
var1	10 / 20	0.515157	4.33%	0.023291 (0.0055)	Accepted
var2	1 / 0.7	0.261935	0.12%	0.000316305 (0.000402)	Kept 1

Var1 alpha 20 passed both safeguards, so its full degree search was rerun with the focused values added. Var2 retained alpha 1 because its local improvement was not meaningful.

Prediction and reproduction

Each final model was fit on its full training set and predicted the matching test rows in their original order. Both output files were checked for exactly one y column, 1,000 finite predictions, and no index column: outputs/BT2024146_pred_var1.csv and outputs/BT2024146_pred_var2.csv. No hidden-test score is reported. Place the five instructor CSVs in data/ as listed in README.md, then run `python src/train_and_predict.py` and `python src/build_report.py`. Code and artifacts: [GitHub repository](#).